



Semih Onay

Education

- 2011–2015 **İstanbul Bilgi University**, Faculty of Arts and Science,
Computer Science (Bachelors Degree - English).
- 2008–2011 **İstanbul Arel University**, Vocational School,
Computer Programming (Vocational - Turkish).

Experience

- July 2020 - *Backend Engineer*, **Vivoo**.
December 2021
 - Migrating to microservices from monolith backend application
 - Designing of the backend and app infrastructure
- December 2021 - *Senior Backend Engineer*, **Kortext**.
2021 -
 - Built and maintain core backend libraries and async job processing infrastructure
Present
 - Own CI/CD pipelines and shared configuration libraries across the platform monorepo
 - Develop event streaming endpoints and Azure-based cloud infrastructure
 - Contribute to digital textbook search and RAG platform serving educational institutions
- August 2019 - *Frontend Engineer*, **Betbull**.
July 2020
 - React Web Application Development
 - Nodejs SSR Web App Development
- September 2018 - *Software Specialist II*, **Etiya**.
2018 -
 - Participating Code reviews as a reviewer
- August 2019
 - Angular-SpringBoot Application Development
 - Drupal CMS Development
- April 2016 - *Software Specialist*, **Setrow**.
August 2018
 - Full-Stack Development
 - Improving existing services
 - Development of API's and Services

Projects

2024–2025	ESP32/M5Stack Firmware	<i>M5Stack Tab5 firmware — MIPI-DSI display debugging, OTA updates, embedded C++</i>
2024	KLP Assignment Coach	<i>AI-powered assignment coaching system — LLM integration, agent workflows</i>

Language Proficiency

English **Advanced**

Writing, reading and speaking

Software Development Skills

Paradigms	Functional, OOP, Reactive, Agent-based	Advanced	
Languages	Python, TypeScript, JavaScript	Advanced	Rust, Elixir
Frameworks	FastAPI, NestJS, Spring Boot		Advanced
Web	React, Next.js	Advanced	Angular
Databases	PostgreSQL, Redis, FalkorDB, MongoDB		Advanced
Cloud	Azure (AKS, Container Apps), Docker, Kubernetes	Advanced	Cognitive Search, Solr
AI/ML	RAG, vector search, OCR, MCP		Advanced
Tools	Git, multi-stage CI/CD pipelines, Jenkins, Jira	Advanced	Security hardening, file processing tooling (Unoserver/LibreOffice)
Testing	pytest, Playwright		Advanced
Other	System design, APIs, microservices		Advanced

Interests

Embedded systems, AI agents, local-first systems, open-source, Turkish legal tech, DIY hardware, radio (TA1SMO), philosophy